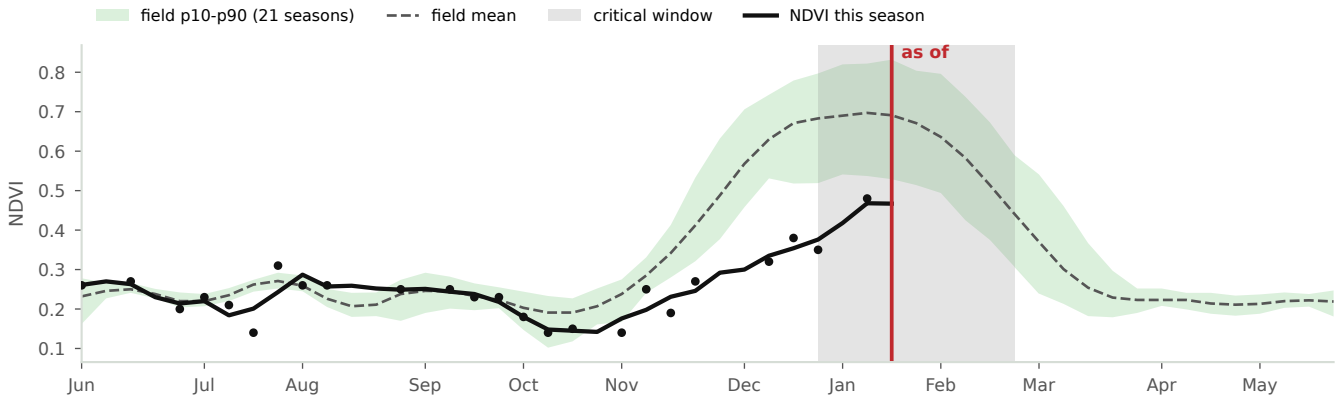
Loan memo: Granja San Pedro S.A.

L047 · Buenos Aires · Norte, Argentina · maize · as of 18 Jan 2023 (week 33)

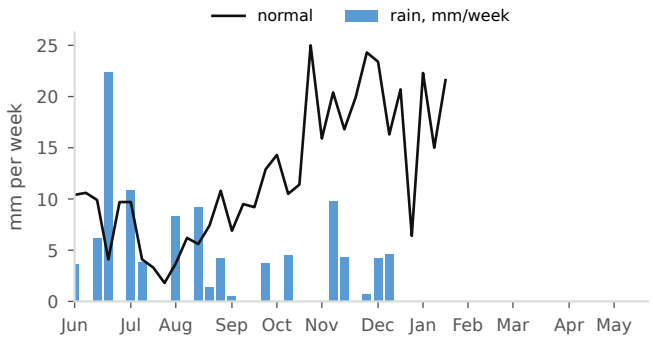
SEVERE score 100/100 · confidence 58% · Stage 2 (critical climate alert (level Severe))

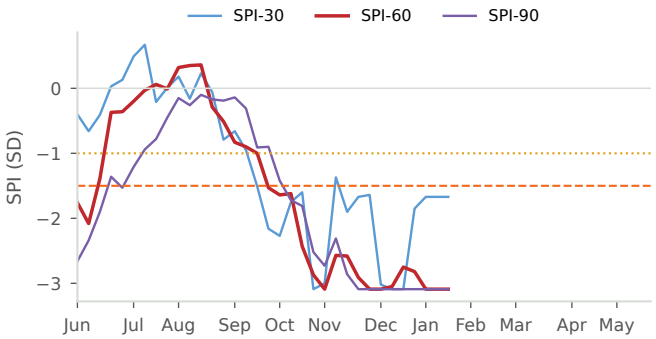
Loan / farmer	**L047 / F045**	Borrower	**Granja San Pedro S.A.**
Product	**Crop input loan**	Amount and rate	**$180,000 at 10.40% p.a.**
Originated / maturity	**18 Sep 2022 / 17 Apr 2023**	Outstanding (EAD)	**$180,000**
Field	**maize, 250 ha; sown 12 Oct 2022**	Crop stage now	**Grain fill (weight 0.80)**
Collateral	**warrant on grain, $184,600**	BCRA / past due	**1 / 0**
Insurance	**parametric, perils: drought; sum insured $185,000; cover 61%; trigger: SPI-60 < -1.5 between flowering and grain fill**	Other-bank debt	**$190,000**

Greenness (NDVI) against this field's own history

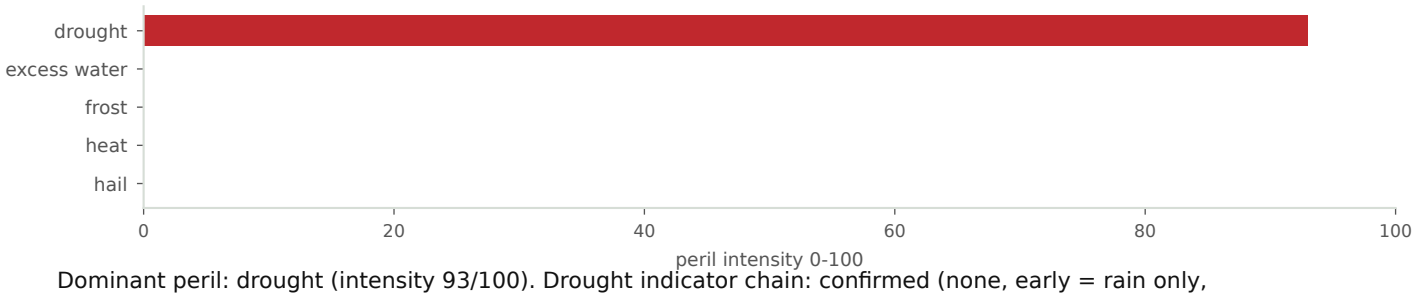


Rainfall and standardised precipitation index





Stress perils, drought cascade and confidence

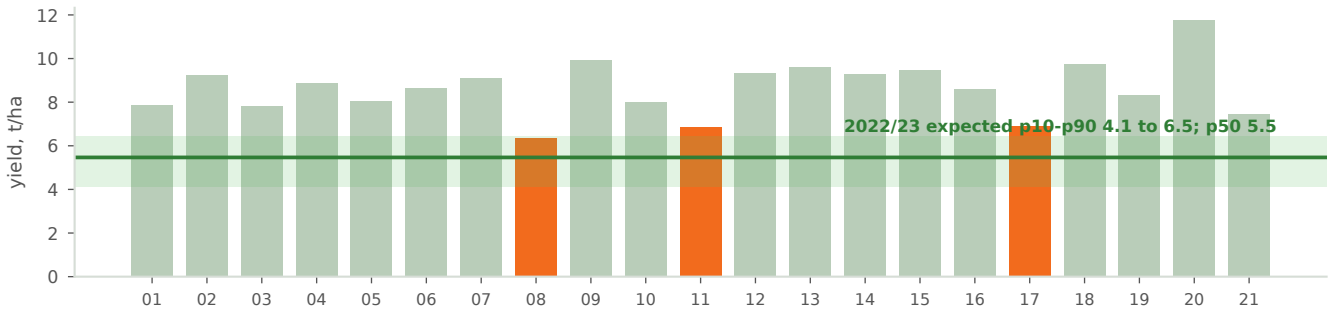


Dominant peril: drought (intensity 93/100). Drought indicator chain: confirmed (none, early = rain only, active = plus soil moisture, confirmed = plus greenness). Confidence 58%: agreement of signals, location quality and share of cloud-free images (0% this week).

Analog seasons and expected yield

Past season	Distance	Yield vs trend	Yield
2017/18	1.15	-24%	6.91 t/ha
2008/09	1.30	-24%	6.35 t/ha
2011/12	1.35	-20%	6.86 t/ha

Expected 2022/23 yield: p50 5.47 t/ha (-34% vs normal 8.27), p10 4.14 (-50%), p90 6.46 (-22%).



Orange bars are the analog seasons; the green band is the p10-p90 range of the regression for this season.

Credit risk: bank baseline against climate-adjusted

Measure	Bank baseline	Climate-adjusted
PD 12-month	9.25% (origination)	21.98% (2.4x)
IFRS 9 stage	performing, Stage 1	Stage 2 (critical climate alert (level Severe))
LGD	44.3%	20.3% (insurance trigger met)
EAD	$180,000	$180,000
ECL base scenario	$7,351	$11,375
ECL adverse scenario (yield p10)	-	$16,808
ECL severe scenario (worst analog)	-	$16,808
ECL probability-weighted	**$7,351**	**$13,548 (+$6,197)**

Explanation

Maize is in grain fill (stage weight 0.8). Drought confirmed: 60-day rain is -3.1 SD from normal (SPI-60), soil moisture is at the 0th percentile and NDVI is 1.9 SD below this field's own history. Expected yield -34% vs normal (range -50% to -22%). Closest past season: 2017/18 (yield -24%). Risk level Severe (score 100, confidence 58%).

Recommendation

File insurance claim and route payout to the loan.

The parametric policy covers drought and its trigger is met; route the payout to loan L047 (61% cover).

Prepared from synthetic data. Numbers are model output for a credit officer to review; no action is taken

automatically.